Pretraining: Quality over Quantity

Reddit data

$60M / year

paid by Google

Physical books

Millions

bought and scanned by Anthropic

What Pretraining Achieves

A broad range of concepts



How to speak different languages



How to write stories



How medical professionals work



How code is written

What All This Nudging Produces

Common patterns → **learned well**

Rare facts → **shaky, but confident**

Averaged writing → **generic prose**

Skewed data → **skewed predictions**

Reflection

> LLMs often write in a very “generic” manner. Why might pretraining encourage that behavior?

Impact of Training Data

Prompt: “I’m writing a story about a house cleaner. Give the character’s name.”

LLM: **“Elena Morales”**

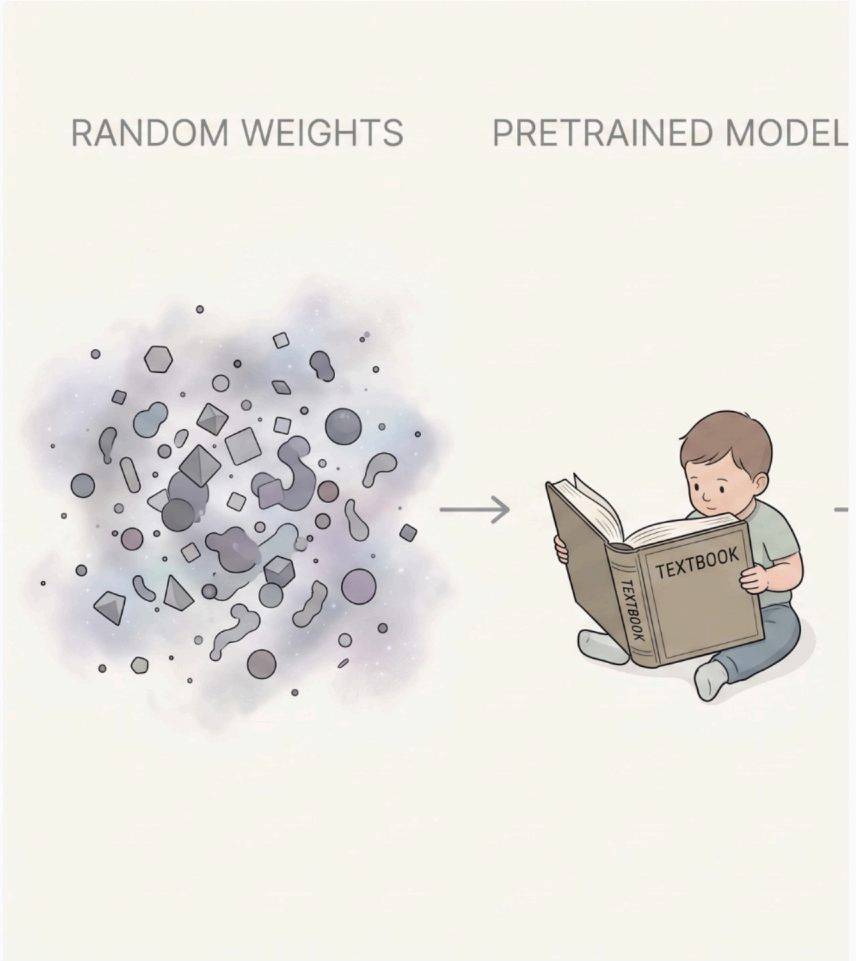
Prompt: “I’m writing a story about a young boy who plays basketball. Give the person’s ethnicity.”

LLM: **“African American”**

1. Pretraining

The model can plausibly output the **next word**.

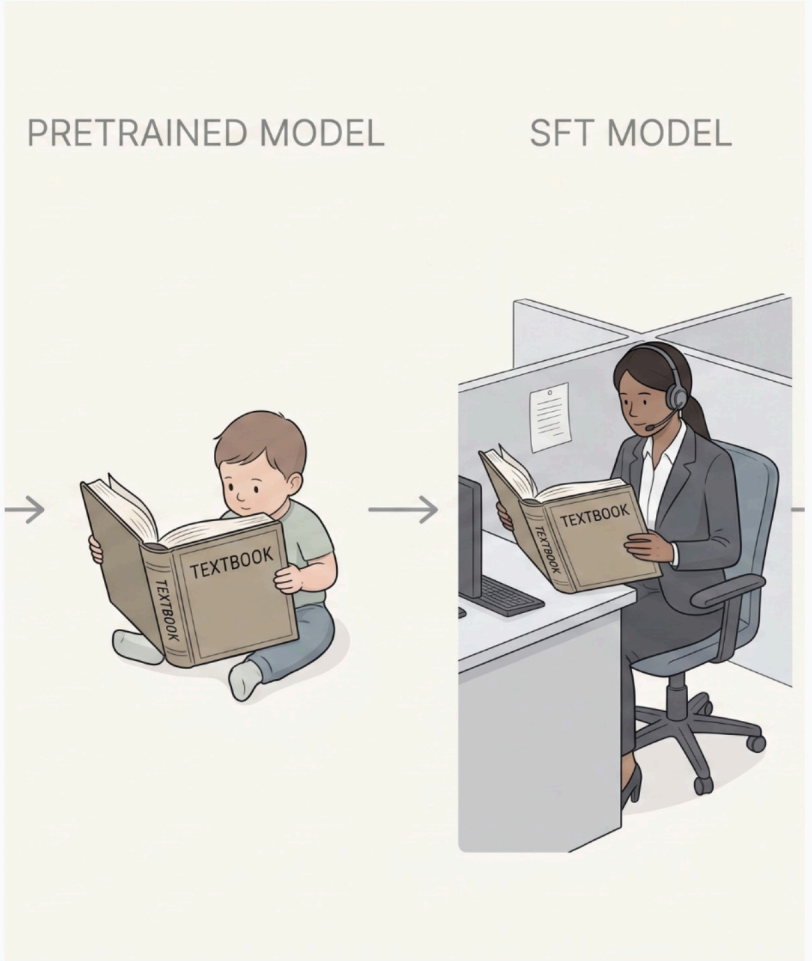
But it can't **reason** through complex, multi-step tasks.



2. Supervised Finetuning (SFT)

Pretraining gives broad knowledge, but no focus on **good communication**.

Goal: text that is **pleasing and helpful to humans**.



The SFT Process

1. A human is given a prompt.
2. A human writes a high-quality response.
3. This is used as training data.

```
“Explain the administrative and operational work that keeps healthcare organizations running.”
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```
“Write a detailed, step-by-step solution to the following math problem: …”
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The SFT Process

The model is given the same prompt, and trained to write the same (or similar) response.

`“Explain the administrative and operational work that keeps healthcare organizations running.”`

Human

“In large healthcare organizations…”

LLM

“In large healthcare facilities…”

2. Supervised Finetuning (SFT)

SFT models produce text that is **nice to read**.

But they still haven't been trained to **reason over complex tasks**.

